Dietary exposure to xenoestrogens in New Zealand.

Continuing evidence of the feminising effects of xenoestrogens on a range of wildlife species increases the need to assess the human health risk of these estrogen mimics. We have estimated the exposure of New Zealand males, females and young men to a range of naturally occurring and synthetic xenoestrogens found in food. Only estrogenic compounds that act by interaction with the estrogen receptor have been included. Theoretical plasma estrogen activity levels were derived from estrogen exposure estimates and estrogenic potency data. Theoretical plasma levels were compared with published data for specific xenoestrogens. There was surprisingly close agreement. Xenoestrogenicity from dietary intake was almost equally attributed to naturally occurring and synthetic xenoestrogens. Relative contributions for a male, for example were isoflavones (genistein and daidzein) (36%) and bisphenol A (34%) with smaller contributions from alkyl phenols (18%) and the flavonoids (phloretin and kaempferol) (12%). It is suggested that dietary xenoestrogens might have a pharmacological effect on New Zealand males and postmenopausal women, but are unlikely to be significant for pre-menopausal women.